

ANIKET SAMANTA

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Professional Summary

AI/ML Engineer and Backend Developer with hands-on experience building RAG-based document intelligence systems, scalable REST APIs, and LLM-integrated applications. Proficient in Spring Boot, Python, FastAPI, and vector databases (FAISS, ChromaDB). Passionate about Generative AI, AI Agents, and production-grade backend engineering.

Technical Skills

Languages	Java, Python
Frameworks	Spring Boot, FastAPI, REST APIs
AI/ML & GenAI	RAG Systems, LLM Integration, Prompt Engineering, Semantic Search, Embedding-Based Retrieval, LangChain, LangGraph
Databases	MySQL, FAISS, ChromaDB
Tools	Docker, Docker Compose, Git & GitHub, Postman, Linux, AWS EC2, React (Vite)
Core Concepts	Data Structures & Algorithms, OOP, System Design Fundamentals

Experience

AI/ML Intern | CSIR – Central Building Research Institute (CBRI) Jan 2026 – May 2026

- Engineered a RAG-based document intelligence chatbot using FAISS & ChromaDB for semantic search over large-scale structured policy data.
- Designed end-to-end retrieval pipeline: document chunking, embedding generation, and similarity-based search using LangChain.
- Developed RESTful APIs using Spring Boot for query handling, user management, and AI pipeline integration.
- Built a React (Vite) frontend interface; implemented RBAC for secure and scalable access management.
- Containerized the full-stack application using Docker & Docker Compose; deployed on AWS EC2 for live, production-accessible hosting.
- Created evaluation pipeline over ~3,000 test cases; achieved 90–95% accuracy on scenario-based queries.
- Analyzed failure categories (PAC, analytical, rule-based) for iterative optimization of retrieval quality.

Data Analyst Trainee | IDEMIA India Foundation (CSR Project) Feb 2024 – Jun 2024

- Completed comprehensive data analyst training: Python, SQL, Excel, data cleaning, EDA, and statistical analysis.
- Built dashboards and reports for data-driven decision-making; skilled in all core data analyst competencies.

Projects

FoodLens — AI-Based Food Ingredient Analysis System

- Designed scalable RESTful backend using Spring Boot with layered architecture and clean separation of concerns.
- Integrated Groq LLM API to generate human-readable nutritional and health insights from ingredient data.
- Built MySQL-based ingredient normalization system; developed ML-based analysis engine with API caching.

Education

Master of Computer Applications (MCA)	Gurukul Kangri Vishwavidyalaya, Haridwar	Aug 2024 – May 2026
Bachelor of Computer Applications (BCA)	Greater Noida Institute of Technology (GNIoT)	Oct 2021 – Jun 2024

Certifications

- Google AI Essentials — Coursera
- Introduction to Generative AI — Google
- Foundations of Project Management — Coursera
- Regression Analysis: Simplify Complex Data Relationships — Coursera

Achievements

- 3rd Place — Hackathon organized by IT Department, Government of Uttarakhand & Gurukul Kangri Deemed University.
- Received Best Coordination Award from college department.
- Participated in Zero Bug — A Bug War problem-solving challenge.